

The He I 1.0833 μm Triplet Optical Depth in Draine (2026): A Factor-of-Three Cross-Check against MoCHII

Abstract

While cross-checking the MoCHII metastable He I 2^3S_1 diagnostic against B. T. Draine’s recent study of resonant scattering of the He I 1.0833 μm triplet in H II regions (2026, ApJ 999, 3; arXiv:2601.11710), I find that the two treatments agree on the metastable population physics but disagree by exactly a factor of three in the line-center scattering optical depth. The paper’s optical-depth normalization (its Eqs. 8, 9, 11, and 12) is three times larger than the standard atomic-physics result for a given 2^3S_1 column density. The discrepancy traces to the upper-level statistical weights listed in the paper’s Table 1, which are $g_u = 3, 9, 15$ where the true fine-structure degeneracies of $2^3P_J^o$ are $2J + 1 = 1, 3, 5$. The printed weights are internally inconsistent with the NIST A - and f -values that the same table cites: those values are mutually consistent *only* with $g_u = 2J + 1$ and the 2^3S_1 lower-level weight $g_\ell = 3$. MoCHII’s τ_0 normalization, the p-winds/Oklopčić & Hirata (2018) exoplanet standard, and the LaRT 1.0833 μm module all use the $2J + 1$ weights and agree with one another. Because H II regions are optically thick to the triplet under either normalization, the paper’s qualitative conclusions (broad multi-peaked profiles, an overall blueshift, and dust-absorption enhancement by resonant trapping) are unaffected, but the quantitative τ_{tot} thresholds and the derived column-density scale should be read a factor of three lower. All numerical claims below were recomputed independently.

1. Context

The He I 1.0833 μm triplet arises from the $2^3S_1 - 2^3P_J^o$ transition and is fed by recombination of He^+ into the triplet system, which cascades into the long-lived metastable 2^3S_1 level. MoCHII computes the 2^3S_1 population as a closed-form local balance in every leaf on the converged state (the Oklopčić & Hirata 2018 balance, hereafter OH18), and derives the 1.0833 μm line-center opacity from it; the design and gate results are recorded in docs/HEI_10833_PLAN.md. Draine (2026) treats the same level as the resonant scatterer in a Monte-Carlo transfer calculation of the 1.0833 μm emission profile emerging from an H II region. Since the two efforts model the same atomic level in overlapping regimes, they form a natural mutual check. The population physics agrees; the optical-depth normalization does not. This memo records the cross-check.

Throughout, 2^3S_1 is the metastable lower level (statistical weight $g_\ell = 2J + 1 = 3$), and $2^3P_J^o$ are the three upper fine-structure levels $J = 0, 1, 2$ (true weights $2J + 1 = 1, 3, 5$). The Doppler broadening parameter is b , quoted in the paper’s fiducial units of 10 km s^{-1} , and $N \equiv N(2^3S_1)$ is the center-to-edge column density of metastable helium.

2. Where the paper and MoCHII agree

The metastable population physics is common to both treatments. Draine’s Eq. (1),

$$n(2^3S_1) = \frac{\alpha_{\text{trip}} n_e n(\text{He}^+)}{A_{ms} + n_e k_d}, \quad (1)$$

is the OH18 balance MoCHII solves, reduced to its two dominant terms: recombination into the triplet system as the source, and radiative decay of 2^3S_1 plus collisional transfer to the singlet system as the sink. Draine notes that 2^3S_1

photoionization ($h\nu > 4.77 \text{ eV}$) and Penning ionization with H^0 are usually subdominant, exactly as MoCHII finds in the dense H II regime. The rate coefficients agree to a few percent:

- $\alpha_{\text{trip}}(9000 \text{ K}) = 2.42 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ (Del Zanna & Storey 2022) vs. MoCHII/OH18 2.28×10^{-13} at the same temperature — within 6%;
- $k_d = 3.3 \times 10^{-8} \text{ cm}^3 \text{ s}^{-1}$ (Bray et al. 2000) vs. MoCHII's $q_{31a} + q_{31b} = 3.2 \times 10^{-8}$ from the same Bray source — within $\sim 3\%$;
- critical density $n_{\text{crit}} = A_{ms}/k_d$: 3855 cm^{-3} (Draine) vs. $\sim 3.9 \times 10^3 \text{ cm}^{-3}$ (MoCHII, 3945) — the small difference following directly from k_d ;
- Draine's Eq. (4) center-to-edge $N(2^3S_1)$ formula reproduces the MoCHII low-density example ($Q_{48} = 10$, $n_e = 110 \text{ cm}^{-3}$, $T_e = 10^4 \text{ K}$; $N \sim 1.4 \times 10^{13} \text{ cm}^{-2}$) to within $\sim 25\%$.

On the population side the two codes are consistent within their input atomic-data spread.

3. The factor-of-three discrepancy

The disagreement is entirely in the mapping from a metastable column density N to a scattering optical depth. Draine's Eqs. (8) and (9) give the center-to-edge line-center optical depth for $2^3S_1 \rightarrow 2^3P_J^o$ absorption as

$$\tau_0(2^3S_1 \rightarrow 2^3P_J^o) = (2J+1) \frac{A_{ul} \lambda_{ul}^3 N}{8\pi^{3/2} b} = 29.2(2J+1) \left(\frac{N}{10^{14} \text{ cm}^{-2}} \right) \left(\frac{10 \text{ km s}^{-1}}{b} \right), \quad (2)$$

and summing the three components (his Eq. 11) yields

$$\tau_{\text{tot}} \equiv \sum_{J=0}^2 \tau_0(2^3S_1 \rightarrow 2^3P_J^o) = 262 \left(\frac{N}{10^{14} \text{ cm}^{-2}} \right) \left(\frac{10 \text{ km s}^{-1}}{b} \right). \quad (3)$$

The standard line-center cross section for a Doppler-broadened line, written either in oscillator-strength or in Einstein-A form,

$$\sigma_0 = \frac{\sqrt{\pi} e^2}{m_e c} f_{\ell u} \frac{\lambda_{ul}}{b} = \frac{g_u}{g_\ell} \frac{A_{ul} \lambda_{ul}^3}{8\pi^{3/2} b}, \quad (4)$$

carries the ratio g_u/g_ℓ . With the true weights $g_u = 2J+1$ and $g_\ell = 3$, and the Table-1 wavelengths and A-values, the three components sum to

$$\tau_{\text{tot}}^{\text{std}} = 87.5 \left(\frac{N}{10^{14} \text{ cm}^{-2}} \right) \left(\frac{10 \text{ km s}^{-1}}{b} \right), \quad (5)$$

which I confirmed to be identical whether computed from the oscillator-strength form or the A-value form of Eq. (4). Component by component (per 10^{14} cm^{-2} , $b = 10 \text{ km s}^{-1}$):

J	standard τ_0 ($g_u = 2J+1$, $g_\ell = 3$)	paper's Eq. (8)	ratio
0	9.72	29.2	3.00
1	29.16	87.5	3.00
2	48.60	145.8	3.00
sum	87.5	262.4	3.00

The ratio is $262.4/87.5 = 3.0000$ to the precision of the input data. The paper's Eq. (8) is the standard $\sigma_0 N$ of Eq. (4) with $g_u \rightarrow (2J+1)$ but with the lower-level weight g_ℓ set to unity rather than to its true value $g_\ell = 3$; the missing denominator is the source of the factor of three.

4. Root cause: the statistical weights

The missing g_ℓ is visible directly in the paper's own Table 1, whose g_u column lists 3, 9, 15 where the fine-structure levels $2^3P_J^o$ have degeneracies $2J+1 = 1, 3, 5$. Table 1 places the two side by side.

Table 1: The He I 1.0833 μm triplet: statistical weights and oscillator strengths. The paper's Table 1 lists $g_u = 3, 9, 15$, which are $3\times$ the true fine-structure degeneracies $2J+1$. The $f_{\ell u}$ values, computed from the tabulated $A_{u\ell}$ with the true $g_u = 2J+1$ and $g_\ell = 3$, reproduce the NIST oscillator strengths (their 1 : 3 : 5 ratio and sum 0.539) exactly.

u	J	Draine g_u	true $2J+1$	$A_{u\ell}$ (s^{-1})	λ_{vac} (μm)	$f_{\ell u}$ (this work = NIST)
$2^3P_0^o$	0	3	1	1.0216×10^7	1.083206	0.05990
$2^3P_1^o$	1	9	3	1.0216×10^7	1.083322	0.17974
$2^3P_2^o$	2	15	5	1.0216×10^7	1.083331	0.29958
sum		27	9			0.539

The most likely origin of 3, 9, 15 is a redundant multiplication by the spin multiplicity $2S+1 = 3$: the fine-structure weight $2J+1$ already partitions the full term degeneracy $(2S+1)(2L+1) = 9$ among the three J levels ($\sum_J (2J+1) = 9$), so multiplying each $2J+1$ again by $2S+1 = 3$ double-counts the triplet spin and yields 3, 9, 15 (sum $27 = 3 \times 9$). In the paper's Eq. (8) this same factor of three appears as the omitted $g_\ell = 3$: using the standard $\sigma_0 = (g_u/g_\ell) \cdots$ with the inflated $g_u = 3(2J+1)$ and the correct $g_\ell = 3$ gives $g_u/g_\ell = (2J+1)$, exactly the bare $(2J+1)$ printed in the paper's Eq. (8). The two readings — “Table 1 g_u inflated by 3” and “ g_ℓ omitted from Eq. 8” — are the same statement.

5. Proof by NIST self-consistency

Independently of any external code, the printed weights are inconsistent with the A - and f -values Table 1 itself cites (NIST ASD; Kramida et al. 2024). The exact relation between an Einstein A and an absorption oscillator strength is

$$g_\ell f_{\ell u} = \frac{m_e c}{8\pi^2 e^2} \lambda_{u\ell}^2 g_u A_{u\ell} \implies f_{\ell u} = \frac{g_u}{g_\ell} \frac{m_e c \lambda_{u\ell}^2 A_{u\ell}}{8\pi^2 e^2}. \quad (6)$$

Substituting the Table-1 $A_{u\ell} = 1.0216 \times 10^7 \text{ s}^{-1}$ with the *true* $g_u = 2J+1$ and $g_\ell = 3$ reproduces the NIST oscillator strengths to five digits:

$$f_{\ell u} = 0.05990, 0.17974, 0.29958 \quad (J = 0, 1, 2),$$

in the ratio 1 : 3 : 5 with sum 0.539 — the last column of Table 1. Had the Table-1 $g_u = 3, 9, 15$ been used in Eq. (6) with $g_\ell = 1$ (the implicit choice behind Eq. (2)), the oscillator strengths would be $f = 0.180, 0.539, 0.899$ (sum 1.62), a factor of three above the cited NIST values in every component and in the sum. The NIST A and f tabulations are therefore

mutually consistent *only* with $g_u = 2J + 1$ and $g_\ell = 3$; the Table-1 g_u column and the τ_0 normalization built on it are the outliers.

6. Independent anchors

Three further independent references use the $2J + 1$ weights and agree with the standard normalization:

1. The oscillator-strength form of σ_0 in Eq. (4), derived from first principles for a Doppler-broadened line, gives $\tau_{\text{tot}}^{\text{std}} = 87.5 \text{ per } 10^{14} \text{ cm}^{-2}$ at $b = 10 \text{ km s}^{-1}$, identical to the A -value form.
2. The p-winds / OH18 exoplanet implementation (p_winds/microphysics.py) uses the same f -values and the same σ_0 ; that opacity has been validated against metastable-helium transit observations.
3. The LaRT 1.0833 μm module (LaRT_v2.00) adopts the same $f = 1 : 3 : 5$ ratio and $A = 1.0216 \times 10^7 \text{ s}^{-1}$. A direct numerical closure confirms the two independent implementations agree: feeding a MoCHII $n(2^3S_1)$ grid (the dense H II example, center-to-edge $N = 1.71 \times 10^{15} \text{ cm}^{-2}$) into LaRT yields $\text{tau_pole} = 1.294 \times 10^3$ (strongest component, center to edge), while MoCHII's blended doublet $\tau_0 = 4.12 \times 10^3$ (full diameter) converts to the same convention as $4.12 \times 10^3 \times \frac{1}{2} \times f_2 / (f_1 + f_2) = 1.288 \times 10^3$ — agreement to 0.5%.

MoCHII's own τ_0 normalization sits in this group. For thermal broadening at $T_e = 10^4 \text{ K}$ ($b_{\text{therm}} = 6.45 \text{ km s}^{-1}$, $v_{\text{turb}} = 0$), MoCHII gives a blended line-center opacity of the $J = 1, 2$ doublet $\sigma_0 = 1.21 \times 10^{-12} \text{ cm}^2$, i.e. $\tau_0 = 1$ at $N(2^3S_1) = 8.3 \times 10^{11} \text{ cm}^{-2}$, and a doublet-to- $J=0$ optical-depth ratio $(f_1 + f_2)/f_0 = 8.00$ — consistent with the f -value ratios of Table 1.

7. Implications

The paper's Monte-Carlo profiles are parameterized by τ_{tot} , and the transfer results as functions of τ_{tot} are unaffected — the issue is only the mapping from physical conditions to τ_{tot} . For a given metastable column density, or a given (Q_{48}, n_3) , the true optical depth is one-third of the paper's quoted value. Table 2 collects the affected coefficients.

Table 2: Optical-depth coefficients: the paper vs. the standard normalization. All quantities scale as $(N/10^{14} \text{ cm}^{-2})(10 \text{ km s}^{-1}/b)$ except Eq. (12), which scales as $\xi^{1/3} Q_{48}^{1/3} n_3^{4/3} / (1 + 0.26n_3)$.

quantity	Draine (2026)	standard / MoCHII	ratio
single-component coeff. (Eq. 9)	$29.2(2J+1)$	$9.72(2J+1)$	3
τ_{tot} coeff. (Eq. 11)	262	87.5	3
τ_{tot} from (Q_{48}, n_3) (Eq. 12)	448	149	3

The paper's contours and example thresholds should be read accordingly: the τ_{tot} values quoted for M51 NE-Strip H II ($\sim 180\text{--}10^3$), M17-B (~ 2000), and NGC 3603 Source E (~ 940) are three times the true values. The qualitative conclusions nonetheless stand. Bright H II regions are optically thick to the 1.0833 μm triplet under either normalization — MoCHII's own τ_0 for representative H II regions ranges from a classical example (doublet $\tau_0 \approx 34$) to a dense example

($\tau_0 \sim 4 \times 10^3$) — so the predicted broad, multi-peaked profiles, the $\sim 14 \text{ km s}^{-1}$ blueshift, and the enhancement of dust absorption by resonant trapping all survive. What changes is the quantitative placement of the transition thresholds and the column-density scale inferred from an observed profile: at fixed physical conditions the resonant-scattering effects set in at three times the τ_{tot} the paper assigns to those conditions.

Because the printed statistical weights are inconsistent with the NIST A - and f -values the paper itself cites, and the standard normalization is confirmed by three independent implementations, the factor of three appears to be a transcription of the upper-level degeneracies in Table 1 rather than a difference of convention — a candidate for an erratum. MoCHII's τ_0 normalization is unaffected and requires no change.

Reproducibility. All numbers in this memo were recomputed from the Table-1 A -values, wavelengths, and CODATA constants; the check script is kept in the session scratchpad.